Fiscal Year:	FY 2017	**Task Last Updated:**	FY 01/30/2017
PI Name:	Alperin, Noam Ph.D.		
Project Title:	Comparing Globe Distortions Following Head-Down Bed Rest to those Measured in Astronauts in Short and Long-Duration Spaceflights		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	33136-2137	**Congressional District:**	24
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2014-15 HERO NNJ14ZSA001N-MIXEDTOPICS. Appendix E: Behavioral Health & Human Health Countermeasures Topics
Start Date:	03/17/2016	**End Date:**	09/06/2017
No. of Post Docs:	1	**No. of PhD Degrees:**	2
No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Allcorn, Aaron	**Contact Phone:**	281.244.8402
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Flight Program:			
Flight Assignment:	NOTE: Project ended early and end date changed to 9/6/2017 (originally 3/16/2018) per A. Allcorn/JSC HRP and NSSC information (Ed., 11/7/17; updated 12/18/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Rittweger, Joern M.D. (German Aerospace (DLR) Deutsches Zentrum Fuer Luft- Und Raumfahrt)		
Grant/Contract No.:	NNX16AG96G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Distortions of the eye globe, flattening of the posterior wall, and protrusion of the optic nerve are key features of the visual impairment intracranial pressure (VIIP) syndrome reported in astronauts following long-duration exposure to microgravity. Therefore, ground-based analog for VIIP should be able to induce similar distortions in humans on Earth. However, the current assessment of globe distortions are done by visual inspection of the MR images and therefore are subjective and lack the sensitivity needed to reliably detect small changes. Our group developed novel enabling technology that automatically quantifies the degree of the globe distortions by MRI. We propose to employ this enabling technology to assess the efficacy of head-down bed rest as ground-based analog for VIIP by detecting and measuring the magnitude of changes in globe flattening, optic nerve protrusion, visual acuity, and other related imaging measures that may be induced in subjects undergoing 30 days head-down bed rest. The magnitude of the changes seen in bed rest will be compared with changes measured in astronauts without and with signs of VIIP. Reference measures in crewmembers without VIIP are from astronauts who flew short-duration missions on the shuttle and the ones with signs of VIIP are from astronauts who flew long-duration missions on the International Space Station (ISS). The data from the short and long duration crewmembers provide valuable references for the magnitude of changes that needs to occur on Earth for head-down bed rest to be considered a relevant analogue for VIIP. A secondary aim is to assess the efficacy of bed rest as a behavioral, health, and performance stressors. Neurocognitive testing, some of which are similar to those performed in the International Space Station (ISS), will be administered to the subjects undergoing the head-down bed rest. Magnitude and rate of decrease in performance will be compared to data available from the ISS to determine the efficacy of the bed-rest model as a ground-based analog for cognitive and behavioral modifiers. Our proposed two-year study is designed to answer the following question posed in the recent Human Exploration Research Opportunities (HERO) solicitation: • Does 30 Days of Head-Down Bed Rest in Humans Combined with Ambient 0.5 % CO2 Induce Changes Related to the Spaceflight Associated Vision Impairment Intracranial Pressure (VIIP) Syndrome and Alter Cognitive Function? [EDITOR'S NOTE (12/18/2017): Project ended early and end date changed to 9/6/2017 (originally 3/16/2018) per A. Allcorn/JSC HRP]
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The research methodologies being develop will elucidate the cause for the VIIP syndrome, and will help identify effective countermeasures to eliminate the development of irreversible ocular damage that is likely to occur in crewmembers during long duration space exploration missions.
Task Progress:	Summary of activities Following the notice of the award, our site started with assembling the resources and research personnel needed for the project, initially scheduled to start in the fall of 2016. Accordingly, research assistants with the required background have been assigned to the project. During the following months, the Principal Investigator (PI) worked with Dr. Lorrie Primeaux, the Analog Science lead of the project, on integrating our research plan with the research protocols of the other PIs. This process and the Institutional Review Board (IRB) approval have been now completed. We then were notified that the start of the project has been delayed to September 2017. Therefore, we used MRI data from a previous bed rest study provided to us by Dr. Joern Rittweger from the DLR, to develop and test some of the analyses techniques that we plan to use for our bed rest study, which is now named VaPER. The primary aim of our project is to determine if the HDT (head down tilt) study is a viable ground analogue for the spaceflight-induced VIIP-related ocular changes. Our work with data collected from astronauts who had signs of VIIP demonstrated that orbital cerebrospinal fluid (CSF)--the CSF around the optic nerves--play a critical role in the development of VIIP (Alperin et al., in press). We applied our automated method to quantify the changes in the orbital CSF volume induced by head down tilt at various tilt angles. The results of this study were presented at the 2017 Human Research Program Investigators' Workshop (HRPIW) meeting. [EDITOR'S NOTE (12/18/2017): Project ended early and end date changed to 9/6/2017 (originally 3/16/2018) per A. Allcorn/JSC HRP]
Bibliography Type:	Description: (Last Updated: 12/18/2017)
Articles in Peer-reviewed Journals	Gerstl L, Schoppe N, Albers L, Ertl-Wagner B, Alperin N, Ehrt O, Pomschar A, Landgraf MN, Heinen F. "Pediatric idiopathic intracranial hypertension - Is the fixed threshold value of elevated LP opening pressure set too high?" Eur J Paediatr Neurol. 2017 Nov;21(6):833-41. Epub 2017 Aug 12. http://dx.doi.org/10.1016/j.ejpn.2017.08.002 ; PubMed PMID: 28838819 , Nov-2017